Local Learning Rules Under Honest Measurement: A Reproducibility-First Benchmark of e-prop, Hebbian, and R-STDP on Consumer Hardware

lzmd-arch*
github.com/lzmd-arch/BioSNN-Plug

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Abstract

[TODO: 120-180 words. Outline: (1) problem – local-rule benchmarks lack honest measurement (budget/protocol/degenerate solutions); (2) what we did – three rule lines + interaction matrix + same-budget BPTT control, consumer GPU; (3) headline numbers – e-prop ALIF converges at 0.8831 (250ep); LIF-vs-ALIF interaction; Critic-corrected R-STDP 237.8 steps; (4) methodological contributions – bit-exact reproduction, retention-joint reporting, alpha-to-zero degeneracy analysis; (5) stance – gaps reported as-is, no cosmetics.]

1 Introduction

1.1 Motivation

1.2 The measurement gap

1.3 Contributions

- **Honest benchmark**: three local-rule lines under one unified protocol (budget/seeds/protocols fully public). [TODO: fill numbers]
- **Interaction finding**: neuron type x learning rate interaction; at default hyperparameters the adaptation term can hurt (same-budget LIF leads ALIF by 1.04pp).
- **Convergence evidence**: e-prop + ALIF converges at 0.8831 (250 epochs), lower edge of the e-prop literature band.
- **Methodology**: bit-exact reproduction, retention-joint reporting, degeneracy analysis, frozen-noise evaluation protocol.
- **Open assets**: all code/configs/scripts open (biosnn-bus on PyPI); comparison table with evidence ledger for 5 sibling projects.

*Independent researcher. Correspondence: [TODO: email].

1.4 Positioning: what this paper is not

2 Related Work

2.1 Local learning rules for spiking networks

2.2 Meta-learned plasticity and arbitration

2.3 SNN–LLM hybrid architectures

2.4 Reproducibility protocols in SNN training

3 Methods

3.1 Systems under test

3.2 Evaluation protocol

3.3 Triviality analysis

3.4 Ethical / boundary statement

4 Experimental Setup

5 Results

5.1 W1: Hebbian feature learning

5.2 W2: e-prop sequential learning

5.2.1 Learning-rate sweep

5.2.2 Neuron type interaction

5.2.3 Budget extension

5.2.4 Comparison with BPTT

5.3 W3: R-STDP with Critic correction

5.4 SHD: an honest negative result

5.5 Resource measurements

6 Discussion

6.1 When do local rules suffice?

6.2 The interaction matters more than the rule

6.3 Limits

7 Conclusion

References

A Full configuration dumps

B Reproduction checklist

C Triviality analysis details